

CyberShield AI

Source Code Documentation

Deployed Website: <https://cybershield-ai-1-5gyk.onrender.com/>

GitHub Repository: <https://github.com/Prasanth4689/CyberShield-AI>

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1. Repository Structure

```
CyberShield-AI/
  ml/
    data_generator.py      # Synthetic log generator (50K events)
    feature_engineering.py # 43 behavioral features pipeline
    baseline_model.py      # Isolation Forest + OC-SVM
    detection_model.py     # XGBoost + LSTM Autoencoder
    classifier.py          # Attack-type classifier + SMOTE
    explainer.py           # SHAP TreeExplainer & NL generator
    evaluate.py            # Metrics, confusion matrix, ROC/PR
    train_pipeline.py      # Training orchestrator
  backend/
    app.py                 # Flask REST API + no-cache static server
    config.py              # Central settings and paths
  dashboard/
    index.html             # Multi-page SPA (Overview/Analytics/Alerts/Model)
    css/styles.css         # Dark/Light theme glassmorphism CSS
    js/api.js              # API client with mock fallback
    js/charts.js           # Chart.js visualizations
    js/app.js              # Tab router, global search, theme, events
  models/                  # Trained model joblib files
  datasets/
    generated/             # Raw synthetic access logs
    processed/             # Features & alerts JSON
  reports/figures/         # Evaluation plot images
  run.py                   # Master CLI (train / serve)
  Procfile                 # Render gunicorn entry point
  render.yaml              # Render deployment manifest
```

2. Code Module Documentation

ml/data_generator.py

Generates 50,000+ access events for 200 entities (150 users, 30 service accounts, 20 edge devices) with 7 injected attack patterns.

ml/feature_engineering.py

FeatureEngineer class extracting 43 behavioral features: temporal, geo-velocity (km/h), behavioral ratios, device fingerprint changes, and historical baselines.

ml/baseline_model.py

BaselineProfiler: Isolation Forest + One-Class SVM ensemble trained on normal traffic.

ml/detection_model.py

AnomalyDetector: XGBoost binary classifier + LSTM Autoencoder. Calculates continuous risk score (0-100).

ml/classifier.py

AttackClassifier: Multi-class XGBoost model with SMOTE oversampling for 7 attack types.

ml/explainer.py

AlertExplainer: SHAP TreeExplainer generating feature contributions and natural language narratives.

backend/app.py

Flask API serving 12 REST endpoints and dashboard static files with no-cache headers.

dashboard/ (index.html, app.js, styles.css, charts.js, api.js)

Multi-page dashboard UI supporting Overview, Analytics, Alerts, and Model Performance views. Features Global Search (Ctrl+K), Light/Dark theme, sliding SHAP details, animated KPIs, Chart.js visualizations, and mobile responsiveness.

3. Deployment & Local Execution

Live URL: <https://cybershield-ai-1-5gyk.onrender.com/>
GitHub Repo: <https://github.com/Prasanth4689/CyberShield-AI>

Local Setup Commands

```
git clone https://github.com/Prasanth4689/CyberShield-AI.git
cd CyberShield-AI
pip install -r requirements.txt
python run.py train    # Trains ML pipeline
python run.py serve    # Starts Flask server on http://localhost:5000
```